

Phase 11 . 8 : Relational Image

The relational image $(R\Downarrow S)$ gives the image of set S under relation R.

Example 1 : Simple relational image

$(R\Downarrow S)$

Example 2 : Relational image with set literal

$(R\Downarrow \{1, 2, 3\})$

Example 3 : Relational image with singleton set

$(parentOf\Downarrow \{\{john\}\})$

Example 4 : Relational image with composition

$((R \circ S)\Downarrow A)$

Example 5 : Relational image with inverse

$(R^\sim\Downarrow S)$

Example 6 : Set comprehension with relational image (Solution 35 pattern)

$\{ p : Person \mid p \in \text{dom } parentOf \bullet (p, (parentOf\Downarrow \{\{p\}\})) \}$

Example 7 : Chained relational images

$((R\Downarrow S)\Downarrow T)$